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GST 103: Philosophy and Logic - Arguments, Fallacies and Clear Thinking

Learn to tell a real argument from noise, and to build one that survives cross-examination

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Philosophy and Logic is where many science and management students switch off - until they meet the objective questions. This pack explains validity, soundness, syllogisms and informal fallacies in plain language, with everyday UNIZIK examples and a fallacy-spotting drill.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Distinguish an argument from a statement, explanation or quarrel.
- Test an argument for validity and soundness without confusing the two.
- Recognise the major informal fallacies in speeches, adverts and social-media arguments.
- Translate simple arguments into standard form and evaluate categorical syllogisms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What philosophy is doing

Philosophy is not a collection of opinions. It is the disciplined practice of asking what a claim means, what it assumes, and what would count as evidence for it.

The main branches you will meet are metaphysics (what exists), epistemology (what knowledge is and how we get it), ethics (how we should act), aesthetics (beauty and art) and logic (the rules of correct reasoning).

For examinations, logic carries the most marks because it is teachable and testable. The other branches matter for definitions and short-answer questions.

- Metaphysics: reality, mind, causation, freedom.

- Epistemology: belief, truth, justification, scepticism.
- Ethics: right action, virtue, consequences, duty.
- Logic: arguments, validity, proof, fallacy.

Arguments: the basic unit

An argument is a set of statements where some, the premises, are offered as support for another, the conclusion.

'I am tired, therefore I will sleep' is an argument. 'I am tired because I slept late' is an explanation. 'You are wrong because you are annoying' is an insult wearing an argument's clothes.

Premises answer 'what is the evidence?' The conclusion answers 'what follows?' Indicator words help: therefore, hence, so, thus and consequently usually introduce conclusions; because, since, given that and for usually introduce premises.

> **Validity versus soundness**
> Validity is about structure: if the premises were true, the conclusion would have to be true. Soundness is validity plus actually true premises. An argument can be valid and unsound, but never sound and invalid.

Three patterns worth memorising

Most exam syllogisms use a small number of forms. Learn the patterns and you can evaluate them quickly.

- Modus ponens: If P then Q. P. Therefore Q. Valid.
- Modus tollens: If P then Q. Not Q. Therefore not P. Valid.
- Affirming the consequent: If P then Q. Q. Therefore P. Invalid - other causes of Q may exist.
- Denying the antecedent: If P then Q. Not P. Therefore not Q. Invalid - Q may still happen for another reason.
- Disjunctive syllogism: Either P or Q. Not P. Therefore Q. Valid when the options are genuinely exclusive.

Categorical syllogisms in plain language

A categorical syllogism has two premises and a conclusion built from three terms: major, minor and middle.

Example: All students write examinations. Ada is a student. Therefore Ada writes examinations. The middle term, 'students', links the other two terms.

Watch the quantifiers: all, no, some, some not. 'Some' means at least one, not most. A conclusion cannot introduce a stronger claim than the premises support.

- All A are B; all B are C; therefore all A are C. Valid.
- No A are B; some C are A; therefore some C are not B. Valid.
- All A are B; some B are C; therefore some A are C. Invalid - the C may be among the Bs that are not A.

Informal fallacies: the ones you will actually meet